

[Pranesh S A 251801234-2025-AIDS](#) ▾**P2****Started on** Monday, 31 August 2026, 10:11 AM**State** Finished**Completed on** Wednesday, 30 September 2026, 11:57 AM**Time taken** 30 days 1 hour**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array of N integer, we have to maximize the sum of $\text{arr}[i] * i$, where i is the index of the element ($i = 0, 1, 2, \dots, N$). Write an algorithm based on Greedy technique with a Complexity $O(n \log n)$.

Input Format:

First line specifies the number of elements- n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int compare(const void *a, const void *b) {
5      long long int x = *(const long long int *)a;
6      long long int y = *(const long long int *)b;
7      if (x < y) return -1;
8      if (x > y) return 1;
9      return 0;
10 }
11
12 int main() {
13     int n;
14     if (scanf("%d", &n) != 1) return 0;
15
16     long long int arr[n];
17     for (int i = 0; i < n; i++) {
18         scanf("%lld", &arr[i]);
19     }
20
21     qsort(arr, n, sizeof(long long int), compare);
22
23     long long int maxSum = 0;
24     for (int i = 0; i < n; i++) {
25         maxSum += arr[i] * i;
26     }
27
28     printf("%lld\n", maxSum);
29
30     return 0;
31 }
```

	Input	Expected	Got	
✓	5 2 5 3 4 0	40	40	✓
✓	10 2 2 2 4 4 3 3 5 5 5	191	191	✓
✓	2 45 3	45	45	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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